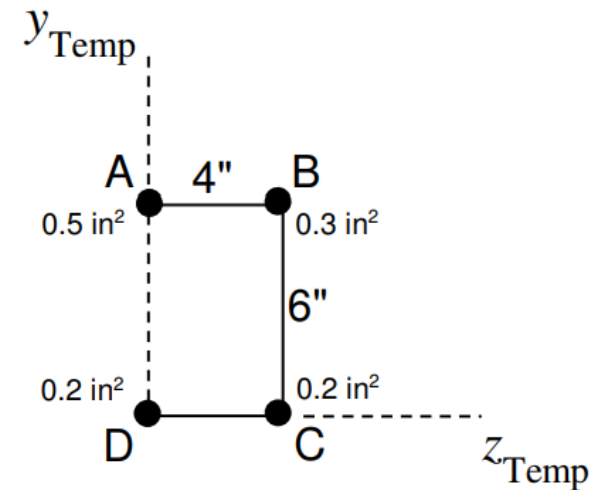


5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- Recall that Part (a) of this example is to find the normal stress in each stringer at the wall of the beam; here are the basic steps to follow:
 - Solve the centroid and place origin of the coordinate frame there
 - Solve inertia values (MOIs and POI)
 - Find value and sign for internal bending moment at the wall
 - Insert these values into the GFF for points A, B, C, and D
- Let's first find the centroid of the cross-section by placing y_{temp} and z_{temp} axes at point D
- Since the webs are thin compared to the effective stringers (and because the idealized webs do not bear normal stress), we will only consider the 4 stringer areas in calculating the centroid
- Label these areas A_A , A_B , etc, and their individual centroid coordinates $\bar{y}_A$, $\bar{y}_B$, etc and $\bar{z}_A$, $\bar{z}_B$, etc



Shear Panel Thicknesses = 0.035"

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- In this idealization, we will treat the cross-section of each effective stringer as a circle whose radius is based on its effective area (here 0.5in^2 , 0.3in^2 , 0.2in^2 , and 0.2in^2)
- Each of these circles is centered at one corner of the cross-section
- Therefore the (y_{temp}, z_{temp}) coordinates of each corner are in fact the (y_{temp}, z_{temp}) coordinates of that stringer's centroid; so the overall centroid of the cross-section is found by:

$$\bar{y} = \frac{\bar{y}_A A_A + \bar{y}_B A_B + \bar{y}_C A_C + \bar{y}_D A_D}{A_A + A_B + A_C + A_D}$$

$$\bar{y} = \frac{(6'')(0.5 \text{ in}^2) + (6'')(0.3 \text{ in}^2) + (0)(0.2 \text{ in}^2) + (0)(0.2 \text{ in}^2)}{\underbrace{0.5 \text{ in}^2 + 0.3 \text{ in}^2 + 0.2 \text{ in}^2 + 0.2 \text{ in}^2}_{1.2 \text{ in}^2}}$$

$$\boxed{\bar{y} = 4''}$$

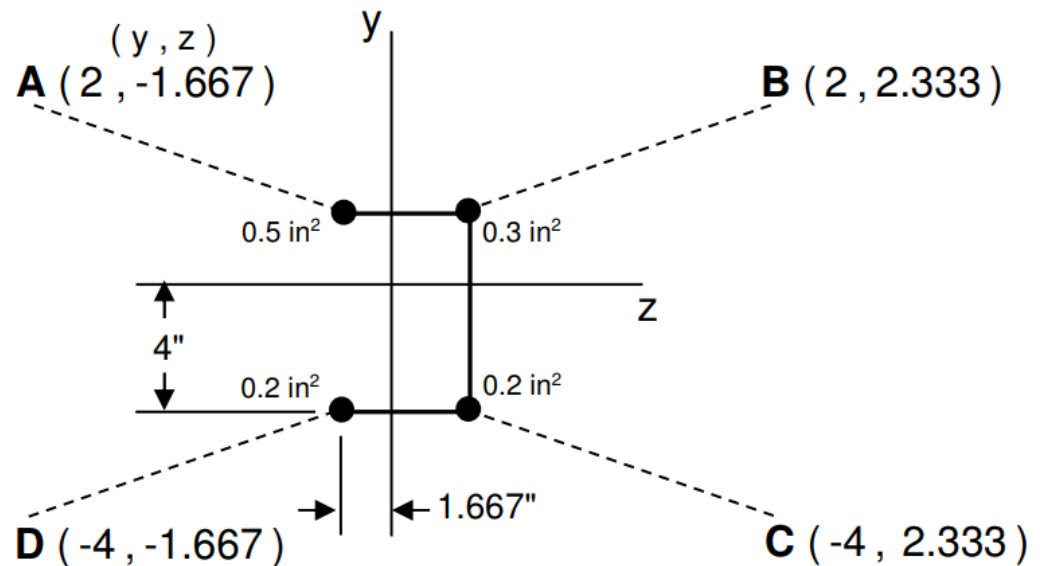
$$\bar{z} = \frac{\bar{z}_A A_A + \bar{z}_B A_B + \bar{z}_C A_C + \bar{z}_D A_D}{A_A + A_B + A_C + A_D}$$

$$\bar{z} = \frac{(0)(0.5 \text{ in}^2) + (4'')(0.3 \text{ in}^2) + (4'')(0.2 \text{ in}^2) + (0)(0.2 \text{ in}^2)}{1.2 \text{ in}^2}$$

$$\boxed{\bar{z} = 1.667''}$$

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- Let's now place the origin of the coordinate frame at the centroid
- Having solved the (y_{temp}, z_{temp}) coordinates of the centroid and knowing the dimensions of the cross-section (4" wide by 6" high), we can easily calculate the centroidal coordinates of points A, B, C, D:



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- Now to solve I_y , I_z , I_{yz}
- For the same reason we neglected the web areas in calculating the centroid, we will also do so in calculating the inertias (i.e. we only consider the 4 stringer areas)
- For Stringer A, the Parallel Axis Theorem yields $(I_A)_y = \overbrace{\frac{\pi(r_A)^4}{4}}^{[I_y \text{ about centroid of A}]} + \overbrace{(A_A)(z_A)^2}^{[\text{Parallel Axis Portion}]}$
- Because Stringer A's effective area is 0.5 in^2 , we have $r_A = \sqrt{\frac{A_A}{\pi}} = 0.3989"$
- So for Stringer A we have $(I_A)_y = \frac{\pi(0.3989")^4}{4} + (0.5 \text{ in}^2)(-1.667")^2$

$$= 0.01989 \text{ in}^4 + 1.389 \text{ in}^4$$

$$(I_A)_y = 1.409 \text{ in}^4$$



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- Notice the first term in this equation (MOI of Stringer A about its own centroidal y axis) is about two orders of magnitude less than the second term (Parallel Axis term)
- In our idealized approach, this will tend to be the case for all of the inertia computations
- Thus, for idealized cross-sections, the inertias will be computed using the following formulas:

$$I_y = \sum_i A_i (z_i)^2$$

$$I_z = \sum_i A_i (y_i)^2$$

$$I_{yz} = \sum_i A_i y_i z_i$$



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- So for this cross-section:

$$I_y = \sum_i A_i (z_i)^2$$

$$= \left\{ \begin{aligned} &(0.5 \text{ in}^2)(-1.667'')^2 + (0.3 \text{ in}^2)(2.333'')^2 \\ &+ (0.2 \text{ in}^2)(2.333'')^2 + (0.2 \text{ in}^2)(-1.667'')^2 \end{aligned} \right\}$$

$$I_y = 4.667 \text{ in}^4$$

$$I_z = \sum_i A_i (y_i)^2$$

$$= \left\{ \begin{aligned} &(0.5 \text{ in}^2)(2'')^2 + (0.3 \text{ in}^2)(2'')^2 \\ &+ (0.2 \text{ in}^2)(-4'')^2 + (0.2 \text{ in}^2)(-4'')^2 \end{aligned} \right\}$$

$$I_z = 9.600 \text{ in}^4$$

$$I_{yz} = \sum_i A_i y_i z_i$$

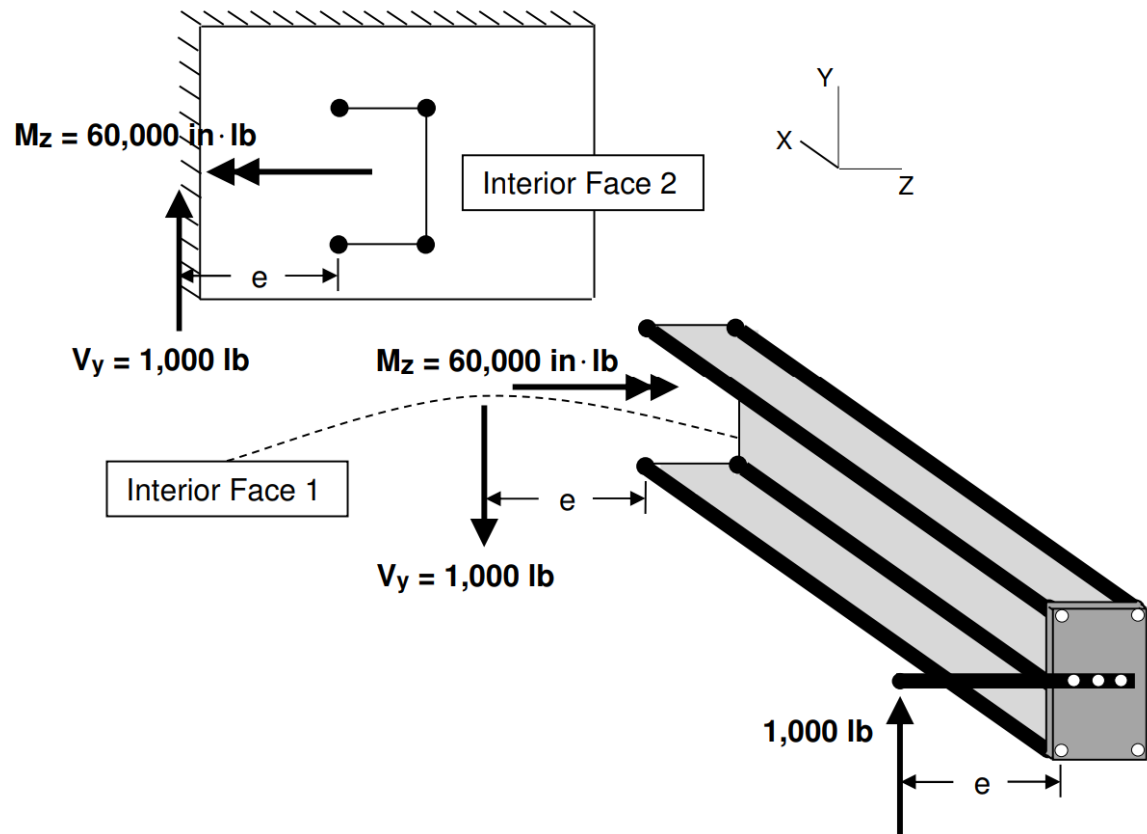
$$= \left\{ \begin{aligned} &(0.5 \text{ in}^2)(-1.667'')(2'') + (0.3 \text{ in}^2)(2.333'')(2'') \\ &+ (0.2 \text{ in}^2)(2.333'')(-4'') + (0.2 \text{ in}^2)(-1.667'')(-4'') \end{aligned} \right\}$$

$$I_{yz} = -0.8000 \text{ in}^4$$



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- The bending moment at the wall due to the 1,000lb applied force is $M_z = 60 \times 1000 = 60,000 \text{ in} \cdot \text{lb}$, depicted here:



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- We can now insert the I_y , I_z , I_{yz} and M_z values into the GFF to obtain:

$$\sigma_x = \frac{-y \left(I_y M_z + I_{yz} \overbrace{M_y}^0 \right) + z \left(I_{yz} M_z + I_z \overbrace{M_y}^0 \right)}{I_y I_z - (I_{yz})^2}$$

$$\sigma_x = \frac{-y \left[(4.667 \text{ in}^4)(+60,000 \text{ in} \cdot \text{lb}) \right] + z \left[(-0.8000 \text{ in}^4)(+60,000 \text{ in} \cdot \text{lb}) \right]}{(4.667 \text{ in}^4)(9.600 \text{ in}^4) - (-0.8000 \text{ in}^4)^2}$$

$$\sigma_x = -(6,341)y - (1,087)z$$



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- Inserting the coordinates of each stringer location (points A, B, C, D) yields:

$$(\sigma_x)_A = -(6,341)(2") - (1,087)(-1.667") \Rightarrow (\sigma_x)_A = -10,870 \text{ psi}$$

- Similarly

$$(\sigma_x)_B = -15,210 \text{ psi}$$

$$(\sigma_x)_C = +22,830 \text{ psi}$$

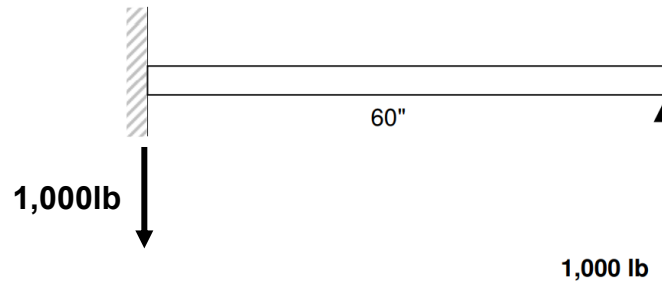
$$(\sigma_x)_D = +27,180 \text{ psi}$$

- Note the + and - signs check out, as we would expect points A and B to be in compression and points C and D to be in tension



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- For Part (b) (calculating shear flow values), we first need to determine the value and sign of the internal shear force in the beam
- Clearly the reaction force at the wall is $V_y = -1,000\text{lb}$



- So if we were to draw a shear force diagram, we would have $V_y = -1,000\text{lb}$ at every location along the beam



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- We now have I_y , I_z , I_{yz} and V_y to insert into the GSFF, resulting in:

$$q_{out} - q_{in} = \frac{\left(I_y V_y - I_{yz} \overset{0}{V_z} \right) \bar{y}' A' + \left(-I_{yz} V_y + I_z \overset{0}{V_z} \right) \bar{z}' A'}{I_y I_z - I_{yz}^2}$$

$$q_{out} - q_{in} = \frac{\left\{ \left[(4.667 \text{ in}^4)(-1,000 \text{ lb}) \right] \bar{y}' A' + \left[-(-0.8000 \text{ in}^4)(-1,000 \text{ lb}) \right] \bar{z}' A' \right\}}{(4.667 \text{ in}^4)(9.600 \text{ in}^4) - (-0.8000 \text{ in}^4)^2}$$

$$q_{out} - q_{in} = \left[-(105.7) \bar{y}' - (18.12) \bar{z}' \right] A'$$

- Given all diagrams so far, we can safely assume that we should calculate shear flow for an interior face directed **away** from the wall (textbook does not specify)

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- In our idealized approach, because the stringers contain σ and the webs contain q , this results in the following assumptions:
 - q will change abruptly (i.e. discontinuously) across each stringer
 - q will remain constant along the web between stringers
- What about the direction of q , i.e. our choice of q_{in} and q_{out} ?
- We can start by assuming a direction and labeling q as follows (note that this figure is consistent with our assumptions above):

